

Francesca Giardine

315-949-8661 | fgiardine@berkeley.edu | [Google Scholar](#) | [Website](#)

EDUCATION

University of California, Berkeley

Doctor of Philosophy in Electrical Engineering and Computer Science

Advised by Dr. Robert Pilawa-Podgurski

Berkeley, CA

Aug. 2020 – May 2026

Smith College

Bachelor of Science in Engineering Science

Bachelor of Arts in Mathematics and Statistics

Summa Cum Laude

Northampton, MA

Aug. 2016 – May 2020

HONORS AND AWARDS

Fellowships

National Science Foundation Graduate Research Fellowship

Tuition and stipend for 3 years of graduate study in STEM.

Sept. 2022 - Sept. 2025

Hearts to Humanity Eternal Graduate Research Grant

\$10,000 grant for independent research.

Sept. 2023 - Sept. 2024

Berkeley EECS Department Fellowship

Semester-long fellowship for select incoming students.

Aug. 2020

Recognitions

Best Paper Award, ECCE Europe 2024

Awarded to 10 of 300 papers presented at ECCE Europe.

September 2024

WiscProf Future Faculty Program Participant

Selective 4-day program sponsored by the University of Wisconsin, Madison providing training for faculty positions in academia.

May 2025

NextProf Nexus Future Faculty Program Participant

Selective program sponsored by the University of Michigan, UC Berkeley, and Georgia Tech giving participants the opportunity to explore and prepare for a faculty position in academia.

August 2024

Adeline Devor Penberthy, Class of 1945, Memorial Prize

Awarded to an undergraduate engineering major for academic excellence in engineering and outstanding contributions toward building a community of learners within the Picker Engineering Program.

May 2020

Honor Societies

Phi Beta Kappa Honor Society

May 2020

Tau Beta Kappa Engineering Honor Society

October 2018

Mu Sigma Rho Statistics Honor Society

May 2020

Sigma Xi Research Honor Society

May 2020

Journal Publications

1. N. C. Brooks, **F. Giardine** and R. C. N. Pilawa-Podgurski, "DC-Link Capacitors for Twice-Line Frequency Power Decoupling: Design-Oriented Figures-of-Merit with Empirical Application," *IEEE Transactions on Power Electronics*, Volume: 39, pp. 6569 - 6573 Issue: 6, June 2024
2. J. Ford, M. Paretti, D. Kotys-Schwartz, S. Howe, C. Gewirtz, J. Deters, T. Mahmud Chowdhury, R. Ott, N. Alvarez, D. Knight, C. Hernandez, L. Rosenbauer, A. Kary, **F. Giardine**. "Transitioning from Capstone Design Courses to Workplaces: A Study of New Engineers' First Three Months." *International Journal of Engineering Education*, June 2019

Peer-Reviewed Conference Publications

1. **F. Giardine**, S. Krishnan, M. Nerenberg* and R. C. N. Pilawa-Podgurski, "An Analytical Model for Common-Mode EMI in Flying Capacitor Multilevel Converters" *2025 IEEE Workshop on Control and Modeling of Power Electronics (COMPEL)* June. 2025
2. E. Krause, N. Biesterfeld, **F. Giardine**, and R. C. N. Pilawa-Podgurski, "Analysis of Steady-State Balancing in the Flying Capacitor Multilevel Converter Considering Capacitor Voltage Ripple" *2025 IEEE Workshop on Control and Modeling of Power Electronics (COMPEL)* June. 2025
3. S. Krishnan, **F. Giardine**, J. Zou, R. Abramson, and R. C. N. Pilawa-Podgurski, "Design and Implementation of a GaN-based, Cascaded Isolated DC-DC Converter for Satellite Applications" *2025 IEEE/AIAA Transportation Electrification Conference & Electric Aircraft Technologies Symposium* June. 2025 (**Best Poster Award**)
4. **F. Giardine**, S. Krishnan, L. Horowitz, and R. C. N. Pilawa-Podgurski, "A Variable Frequency Technique for EMI and Efficiency Improvements in High-Level Count Flying Capacitor Multilevel Converters" *2025 IEEE Applied Power Electronics Conference*, Mar. 2025
5. **F. Giardine**, Y. Wu and R. C. N. Pilawa-Podgurski, "A Variable Switching Frequency Control Technique for DC-AC Flying Capacitor Multilevel Converters to Improve Efficiency and Inductor Utilization" *2024 IEEE Energy Conversion Congress and Exposition*, Oct. 2024
6. **F. Giardine**, K. Fernandez and R. C. N. Pilawa-Podgurski, "A Two-Stage Non-Isolated Hybrid Switched-Capacitor Microinverter Utilizing a Fixed-Ratio Resonant DC-DC Stage with Startup Functionality and Flying Capacitor Multilevel DC-AC Stage" *2024 IEEE Energy Conversion Congress and Exposition - Europe*, Sept. 2024 (**Best Paper Award**)
7. **F. Giardine**, N. C. Brooks, K. Fernandez and R. C. N. Pilawa-Podgurski, "Utilizing Harmonic Injection to Reduce Energy Storage and Required Capacitance in an Active Series-Stacked Energy Buffer for Single-Phase Systems," *2022 IEEE 23rd Workshop on Control and Modeling for Power Electronics (COMPEL)*, June 2022
8. C. Gerwitz, **F. Giardine**, R. Ott, A. Kary. "Women's Unique Challenges in the Transitions to Engineering Work." *2020 ASEE Annual Conference & Exposition*, June 2020

9. S. Howe, M. Paretti, R. Ott, J. Deters, C. Gewirtz, **F. Giardine**, C. Hernandez, A. Kary. "Women's Experiences in the Transition from Capstone Design Courses to Engineering Workplaces." *2019 ASEE Annual Conference & Exposition*, June 2019

*denotes undergraduate student

TEACHING

University of California, Berkeley EECS

EE 108: Introduction to Electric Power and Renewable Energy *Spring 2024, Spring 2025*
Certificate in Teaching and Learning in Higher Education *Expected, Fall 2025*

Smith College

Engineering Master Tutor *Spring 2019 - Spring 2020*
General Quantitative Tutor *Fall 2018*

RESEARCH EXPERIENCE

Pilawa Research Group

University of California, Berkeley *Aug. 2020 - Present*
Berkeley, CA

- Developing advanced hybrid switched capacitor converters for use in DC-AC grid-connected systems.
- Validation of fundamental converter and control behavior in simulation and hardware.

Power Electronic Device Characterization Internship

National Renewable Energy Laboratory *June 2023 - Aug. 2023*
Golden, CO

- Designed a test setup for switch characterization for power electronics.
- Performed simulation, printed circuit board layout, and investigated advanced sensing techniques to achieve high-bandwidth performance.

Researcher in the Renewable and Appropriate Energy Lab

University of California, Berkeley *July 2020 - Aug. 2020*
Berkeley, CA

- Continued research from the SUPERB REU under Professor Daniel Kammen in the Energy Resources Group.
- Performed statistical and spatial analyses in Python and GIS to isolate candidate communities for microgrid development

SUPERB Research Experience for Undergraduates

University of California, Berkeley *June 2019 - Aug. 2019*
Berkeley, CA

- Analyzed geospatial data in RStudio and GIS to classify community types in California with emphasis on environmental predictors.
- Selected as a representative to present research at NSF REU Research Symposium in October 2019.

Capstone to Work Research Team

Smith College *Sept. 2018-May 2020*
Northampton, MA

- Reviewed qualitative and quantitative data from a large pool of first year engineering employees to analyze the impacts of the engineering capstone experiences on employment after college.

SERVICE AND LEADERSHIP

University of California, Berkeley

Berkeley EECS Graduate Admissions Student Reader

2023, 2024

IEEE Power and Energy Society

Jan. 2022 - Present

- 2023 Chair
- 2022 Vice Chair

EECS Graduate Student Association Committee Member

Sept. 2021 - Sept. 2022

Smith College

Tau Beta Kappa Engineering Honor Society (Co-President)

Sept. 2019 - May 2020

Writing and Public Discourse Committee Student Liaison

Sept. 2018 - May 2020

Outdoor Orientation Leader

Fall 2019

MENTORING

Jack Armstrong (Undergraduate)

May 2025 - Present

Hysteretic Control for High-Level Count FCML Converters

Yayun Zhao (Undergraduate)

October 2024 - Present

Design of a High-Performance Totem-Pole PFC Circuit

Nikhil Ograin (Undergraduate)

September 2024 - May 2025

A Resonant Flying Capacitor Multilevel Inverter